Monthly Electricity Bill Prediction

Predicting monthly household electricity bill price using regression

TEAM 10

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REPOSITORY

REPOSITORY LINK

https://github.com/2026dab1292-maker/Electricity-Bill-Estimator/blob/main/README.md

Public repository

Code · data · README

requirements · slides · video link

Problem and data

WHAT AND WHY

Electricity bill depend upon various factors so it varies for different households.

This model helps to find relationship between that factors and bill amount.

DATASET

Source:https://www.kaggle.com/datasets/gireeshs/household-monthly-electricity-bill

1000 households × 10 columns

No missing values · No duplicate rows

COLUMNS

Num_rooms,num_people,housearea,is_ac,is_tv,is_flat,ave_monthly_income,num_children,is_urban,amount_paid

Approach

1 DATA

Household_energy_bill_data.csv

2 CLEAN

Num_rooms,num_people,housearea,is_ac,is_tv,is_flat,ave_monthly_income,num_children,is_urban,amount_paid

3 PREPARE

80/20 split, scale, one-hot encode

4 MODEL

Linear Regression (scikit-learn)

5 OUTPUT

Predicted bill (₹) and metrics

KEY STEPS

- Predict bill (₹) directly with Linear Regression.
- Fit numerical scaling on training data only.
- One-hot encode categorical features.

LIBRARIES USED

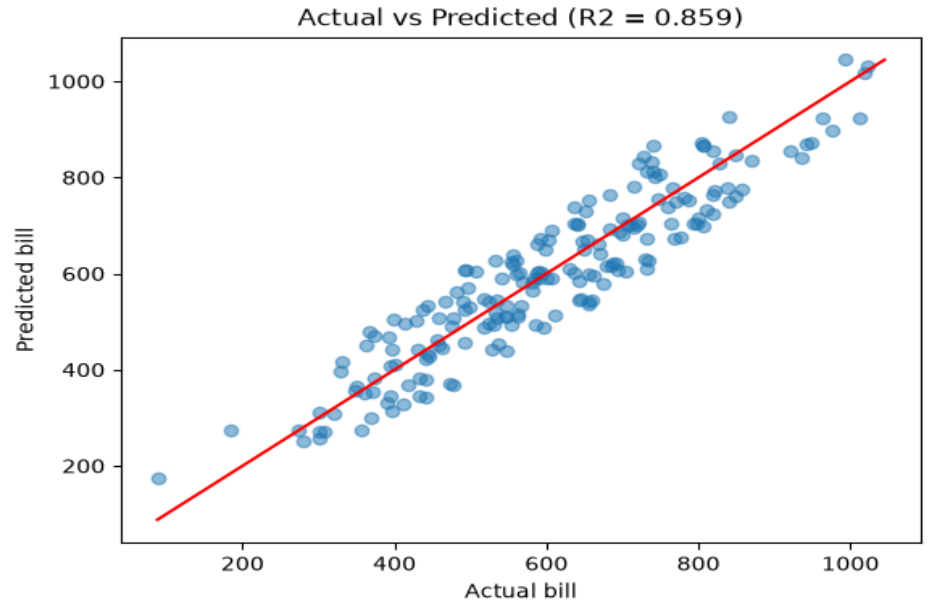
pandas · NumPy · scikit-learn
matplotlib · seaborn · Jupyter

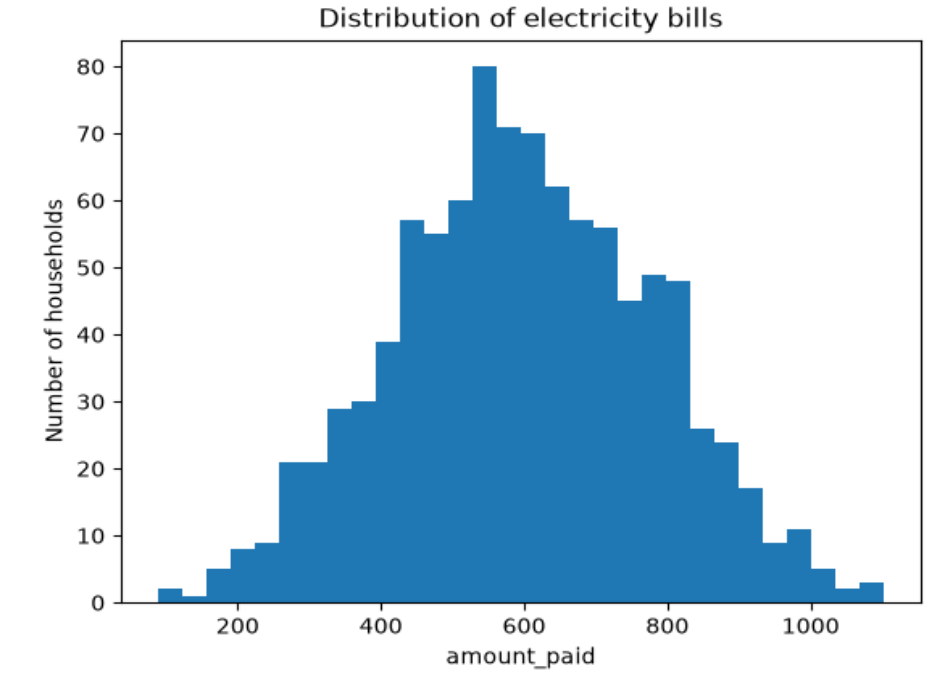
Results

R^2 SCORE (higher is better)

R^2 : 0.859 (TEST)

TEST METRIC	VALUE
R^2	**0.859**
MAE	**55.61rs**
RMSE	**65.67rs**





Learnings and AI use

WHAT WE FOUND

Linear Regression explains about 85.9% of electricity bill variation ($R^2 \approx 0.859$). Remaining 14.1% is not explained by model.

The RMSE is higher than MAE , suggesting that some larger prediction errors may be present.

LIMITATIONS AND NEXT STEPS

LIMITS : If the training dataset contains limited household records ,the model may not perform equally well for all households.

NEXT : use newer data and compare other models, such as Random Forest .Increase dataset size which means collect electricity bills from more households.

AI-USE DISCLOSURE

Claude was used for learning, setup guidance, execution support, and understanding the README format. It was also used to help make the presentation; most of the theoretical content was written by the PPT lead.

ROLES

DIVYDAKSH TIWARI : DOCUMENTATION LEAD , data cleaning

KANNURU NARAYANA PRAYOG : PPT LEAD , data analysis

ASHUTOSH KUMAR : EXECUTION LEAD , training the data

GOKUL GARG : VIDEO LEAD , testing the data